



DEVELOPERS LEVEL · CAPSTONE

# Problem-Solver App

Student Workbook

Design thinking + your full JavaScript toolkit — build a tool for a real problem.

Name: \_\_\_\_\_

## THEORY

# Design Thinking + Your JS Toolkit

You now have a full toolkit: **variables**, **functions**, **if/else**, **arrays**, and **events**. **Design thinking** puts them to work: find a real problem, know your **user**, plan a solution, build a **prototype**, then test and **debug**. This capstone integrates all four semesters.

Discuss: what is a real problem in your day that a small app could help with?

## Frame it

Write the problem and who it is for.

### The problem:

### The user (who it helps):

#### Code we use:

variable   function   if / else   array   onclick   console.log

## Words we are learning

- **Design thinking** — understand a problem, plan, build, test, improve.
- **User** — the person your tool is for.
- **Prototype** — a first working version to test.
- **Function** — the reusable steps your app runs.
- **Debug** — find and fix what does not work.

**Name:** \_\_\_\_\_ **Date:** \_\_\_\_\_

## Find A Real Problem

Choose a real problem you could solve with code. Answer the questions.

**The problem I will solve:**

---

---

**Who is the user, and why do they need it?**

---

---

**Name:** \_\_\_\_\_ **Date:** \_\_\_\_\_

# Plan The Solution

Pick your JS tools and sketch the app.

**Which tools will I use? (circle)**

variables

functions

if/else

arrays

events

sketch your app screen

**Name:** \_\_\_\_\_ **Date:** \_\_\_\_\_

## Build & Trace

Write the main idea of your app in pseudocode (plain-language steps).

```
function main() {
```

```
  // step 1: _____
```

```
  // step 2: _____
```

```
  // step 3: _____
```

```
}
```

**What will you console.log to test it?**

**Name:** \_\_\_\_\_ **Date:** \_\_\_\_\_

## Test & Present

Test your prototype, then plan your presentation.

- ☐ It solves the real problem I chose
- ☐ It uses 3 or more JS tools
- ☐ I found and fixed at least one bug
- ☐ A user could actually use it

**My 30-second pitch:**

---

---

Name: \_\_\_\_\_ Date: \_\_\_\_\_

**PROJECT**

# Problem-Solver App — Plan

Your capstone: pick a real problem and build a JavaScript tool for it, using your full toolkit. Passing graduates you to Engineers.

## 1. The problem and user:

## 2. My tools (circle):

variable

function

if/else

array

event

## 3. Main function steps (pseudocode):

```
function main() { _____ }
```

## 4. How I will test it: